Global Issues

Who is in charge when a problem crosses borders?



Warm-up

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Which of these global issues are regulated by an international organization?

- Pandemics
- Air travel
- The internet
- Mail
- Nuclear technology
- Artificial intelligence

Five out of six

Pandemics	→	**WHO**
Air travel	→	**ICAO**
The internet	→	**ITU** (partly)
Mail	→	**UPU**
Nuclear technology	→	**IAEA**
Artificial intelligence	→	**?**

All five are part of the United Nations system

What is a UN specialized agency?

- An independent organization linked to the UN by treaty
- Its own members, budget and leadership
- Countries choose to join, and can leave
- It sets shared rules, but has no police and no army: it depends on countries to follow them

Pandemics

The problem: pandemics

- A virus does not stop at a border
- COVID-19 reached almost every country within three months
- More than 7 million reported deaths worldwide
- What happens if every country keeps its data to itself?

Source: WHO COVID-19 dashboard.

World Health Organization (WHO)

- **Headquarters:** Geneva, Switzerland
- **Founded:** 7 April 1948 (now World Health Day)
- **Members:** 194 countries
- **Goal:** "the attainment by all peoples of the highest possible level of health"
- **US connection:** the US announced its withdrawal in January 2025 and says it left in January 2026

Source: WHO Constitution, Article 1; HHS, Jan. 2026.

What does that mean?

- Coordinates the global response to outbreaks: collects data, declares emergencies, issues guidance
- Sets health standards that countries can adopt
  - Example (2025): “use of lower-sodium salt substitutes”
- Runs long-term campaigns, such as **MPOWER** against tobacco

Source: WHO, *Use of lower-sodium salt substitutes* (Jan. 2025).

MPOWER: six steps against tobacco

M **Monitor** tobacco use and prevention policies

P **Protect** people from tobacco smoke

O **Offer** help to quit tobacco use

W **Warn** about the dangers of tobacco

E **Enforce** bans on tobacco advertising, promotion and sponsorship

R **Raise** taxes on tobacco

Source: WHO, MPOWER (introduced 2008).

Air Travel

The problem: air travel

- More than 100,000 flights take off every day
- A plane built in the US, flown by an Austrian airline, lands in Japan. On the way it crosses about ten countries
- Which country's rules apply? What language does the pilot speak to the tower?



Austria → Hungary → Romania → Turkey → Georgia → Azerbaijan → Kazakhstan → China → South Korea → Japan

Source: ICAO; Austrian Airlines. Typical route, varies daily. Map: TUBS, CC BY-SA 3.0 (modified).

International Civil Aviation Organization (ICAO)

- **Headquarters:** Montreal, Canada
- **Founded:** 4 April 1947
- **Members:** 193 countries
- **What it does:** sets shared standards for safety, pilot licences, airports and air traffic rules
- **US connection:** created by the Chicago Convention, signed in Chicago in 1944

Source: ICAO.

ICAO standards you have already met

- English: every pilot and controller on international flights must pass an English test (ICAO “Level 4”)
- The alphabet: Alfa, Bravo, Charlie . . . so that “B” and “D” are never confused over the radio
- Your passport: the same size in every country, with two lines of machine-readable code at the bottom
- Black boxes: every airliner must carry flight recorders, and crash reports are shared with all countries
- Airport codes: Vienna is **LOWW** to every pilot in the world

What if a country doesn't follow the rules?

- ICAO has no power to punish anyone. It can only inspect and publish the results
- But other countries can close their airspace to airlines they don't trust
- Example: in 2020, Pakistan found that almost a third of its pilots had fake or suspicious licences
- The EU banned Pakistan's national airline for more than four years (2020–2024)

Source: EASA, Nov. 2024.

The Internet

The problem: staying connected

- Radio signals and satellites ignore borders
- Your phone works when you land in another country
- Every country has its own calling code: $+43$ Austria, $+1$ USA
- Who decides who gets which frequency, and which number?

International Telecommunication Union (ITU)

- **Headquarters:** Geneva, Switzerland
- **Founded:** 17 May 1865, as the International *Telegraph* Union: the oldest UN agency
- **Members:** 194 countries
- **What it does:** divides up radio frequencies and satellite orbits, sets technical standards, assigns country calling codes
- **US connection:** its Secretary-General, Doreen Bogdan-Martin, is American

Source: ITU.

The ITU in practice

- Your laptop finds Wi-Fi in Vienna, New York and Tokyo, because Wi-Fi uses the same frequencies everywhere
- A radio station in Vienna and one in Bratislava (60 km away) don't drown each other out
- Every satellite, including Starlink's, has to be registered with the ITU
- **But:** it does *not* run the internet. Web addresses are managed by ICANN, a non-profit based in California

Source: ITU; ICANN.

Mail

The problem: sending a letter abroad

- You buy an Austrian stamp and post a letter to New York
- The Austrian post gets your money. The US post does the delivery
- Before 1874: a separate treaty between every pair of countries, and a different price for every route
- Who pays the American mail carrier?

Universal Postal Union (UPU)

- **Headquarters:** Bern, Switzerland
- **Founded:** 9 October 1874 (now World Post Day)
- **Members:** 192 countries
- **What it does:** makes the world "a single postal territory." Three rules:
  - Deliver it like your own: every member must deliver foreign mail as if it were domestic mail
  - Freedom of transit: every member must pass on mail that is only travelling through
  - Terminal dues: the sending country pays the delivering country a fixed fee, set by the UPU
- **US connection:** in 2018 the US announced it would leave, saying the fees made it too cheap to ship parcels from China. It stayed after the rules were changed in 2019

Source: UPU Constitution, Article 1; Universal Postal Convention.

Nuclear Technology

The problem: nuclear technology

- The same technology makes electricity and bombs
- Nine countries have nuclear weapons: about 12,000 warheads
- A country says its nuclear program is only for energy. How does anyone know?

Source: SIPRI Yearbook 2025.

International Atomic Energy Agency (IAEA)

- **Headquarters:** Vienna, Austria
- **Founded:** 29 July 1957
- **Members:** about 180 countries
- **What it does:** helps countries use nuclear technology peacefully, and sends inspectors to check that it stays peaceful
- Nobel Peace Prize, 2005
- **US connection:** proposed by President Eisenhower in his "Atoms for Peace" speech at the UN, 1953



Vienna International Centre ("UNO City"). Photo: Dimitry Anikin, CC0

Source: IAEA.

Artificial Intelligence

The problem: artificial intelligence

- Built by a handful of companies, mostly in the US and China
- Used in almost every country
- Rules so far are a patchwork: the EU has its AI Act (2024); most countries have little or nothing
- New risks keep appearing: deepfakes, AI-designed bioweapons, ...

Create your own agency

- **Should there be a "WHO for AI"?**
- Which of today's organizations would be the best model?
- Would the US and China join?